

ALF basics: Overview

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ALF workshop Würzburg

February 14-18, 2022

ALF basics: Introduction

Lectures series on the programming details of working with ALF: ■

- Overview *This Talk*
- Building blocks of a Model
 - Lattice *Following Talk*
 - Observables *Tue. 13:00-13:30*
 - Interactions *Tue. 16:30-17:00*
 - Hopping *Wed. 13:00-13:30*
- ⇒ Building your own Model/Hamiltonian . *Wed. 16:30-17:30*
- Advanced pyALF *Thu. 13:00-13:30*

Outline

Folder overview

Folder overview: Miscellaneous

Folder overview: Monte Carlo code

Folder Prog/

Code structure

Monte Carlo core vs Hamiltonians

Hamiltonian structure

main.F90 structure

Folder overview: Miscellaneous

```

ALF/
├── Documentation/
│   └── doc.pdf.....Documentation
├── Scripts_and_Parameters_files/.....Sample scripts
│   └── Start/.....Files needed for ALF run
│       ├── parameters.....Sample parameter file
│       ├── seeds.....Random number seeds
│       └── out_to_in.sh.....For continuing a run
├── testsuite/.....Suite of unit tests
└── Mathematica/.....Mathematica notebooks
                        for Wick decomposition

```

Folder overview: Monte Carlo code

```
ALF/
├── Analysis/.....Error analysis programs
├── HDF5/.....HDF5 library, automatically
                    downloaded and compiled
├── Libraries/.....General routines, e.g. Lattice
├── Prog/.....Monte Carlo core and Hamiltonians
├── configure.sh.....Sets environment variables
└── Makefile.....Needed by make
```

Folder Prog/

```

ALF/
├── Prog/..... Monte Carlo core and Hamiltonians
│   ├── Hamiltonians.list..... Hamiltonian names
│   ├── Hamiltonian_main_mod.F90.....
│   │           Interface between Hamiltonians
│   │           and Monte Carlo core
│   ├── Hamiltonians/
│   │   ├── Hamiltonian_MyModel1_smod.F90
│   │   ├── Hamiltonian_MyModel1_read_write_parameters.F90.
│   │   │           Automatically generated file
│   │   ├── Hamiltonian_MyModel2_smod.F90
│   │   ├── Hamiltonian_MyModel2_read_write_parameters.F90.
│   │   │           Automatically generated file
│   ├── main.F90..... Main program file
│   └── ALF.out..... Executable

```

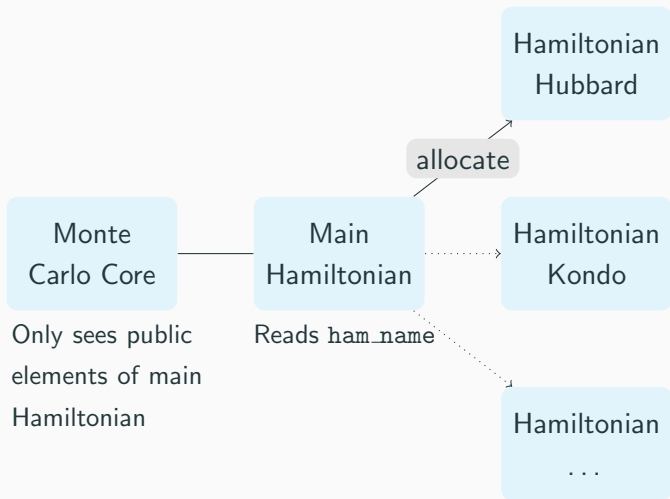
Folder Prog/

```

ALF/
├── Prog/ ..... Monte Carlo core and Hamiltonians
│   ├── Hamiltonians.list ..... Hamiltonian names
│   ├── Hamiltonian_main_mod.F90 .....
│   │   └── Interface between Hamiltonians
│   │       and Monte Carlo core
│   ├── Hamiltonians/
│   │   ├── Hamiltonian_MyModel1_smod.F90
│   │   ├── Hamiltonian_MyModel1_read_write_parameters.F90.
│   │   │   └── Automatically generated file
│   │   ├── Hamiltonian_MyModel2_smod.F90
│   │   └── Hamiltonian_MyModel2_read_write_parameters.F90.
│   │       └── Automatically generated file
│   ├── main.F90 ..... Main program file
│   └── ALF.out ..... Executable

```

Monte Carlo core vs Hamiltonians



Submodules of main Ham

Hamiltonian structure

Peek into `Hamiltonian_main_mod.F90`

main.F90: Initialization

```

read QMC parameters    ▷ From file parameters

call Alloc_Ham          ▷ Select Hamiltonian based on ham_name in
                        parameters

call ham%Ham_Set        ▷ Set the Hamiltonian

call Nsigma%in          ▷ Read in an auxiliary-field configuration or
                        generate it randomly
                        Caveat: To restart simulation rename
                        confout_* to confin_*

for n = L_Trotter to 1 do ▷ Fill the storage needed for the first actual
                        MC sweep

    call Wrapul          ▷ Compute propagation matrices and store
                        them at stabilization points

end for

```

main.F90: Monte Carlo Loop

```

for  $n_{bc} = 1$  to  $N_{Bin}$  do                                ▷ Loop over bins.
  for  $n_{sw} = 1$  to  $N_{Sweep}$  do                                ▷ Loop over sweeps.
    if(Tempering) call Exchange_Step
    if(Global moves) call Global_Updates
    if(Langevin) call Langevin_update
    if Sequential then
      Sweep up
      Sweep down
      call Tau_m                                ▷ MEASURE time-displaced observables.
    end if
  end for (Sweeps)
  call ham%Pr_obs                                ▷ Write measurement for current bin to disk.
  call Nsigma%out                                ▷ Write auxiliary field configuration to disk.
end for (Bins)

```

main.F90: Sequential sweep up

```

for  $n_\tau = 1$  to  $L_{\text{Trotter}}$  do
  call Wrapup                                ▷ PROPAGATE Green function from
                                               $n_\tau - 1$  to  $n_\tau$  via sequential single spin
                                              flip updates.
  if  $n_\tau ==$  stabilization point then      ▷ Do a stabilization step.
    call Wrapup
    call CGR
    call Control_PrecisionG
  end if

  if  $n_\tau \in [\text{Lobs\_st}, \text{Lobs\_en}]$  then
    call ham%Obser    ▷ MEASURE the equal-time observables.
                      Calls ObserT.
  end if
end for

```